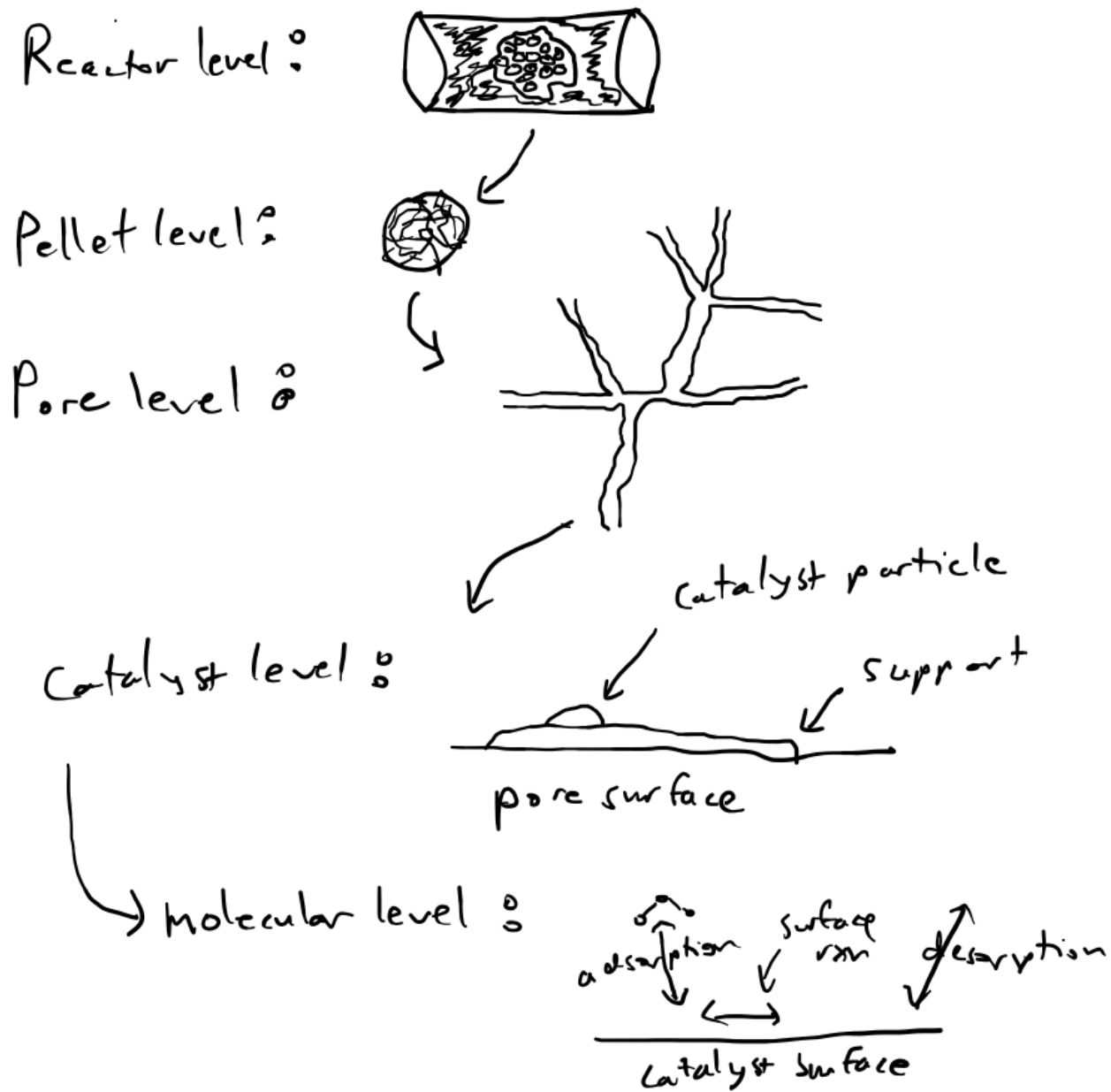
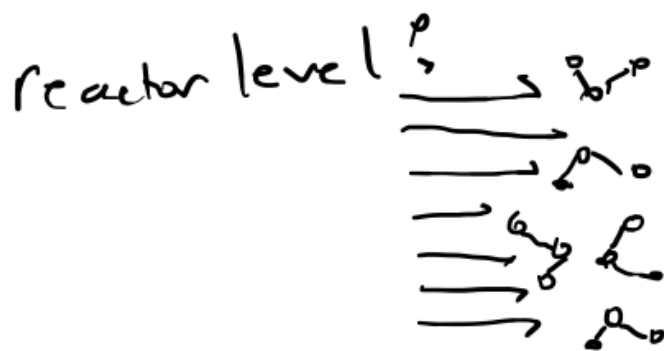


14 Catalytic Reactors



- from the perspective of a gas molecule $\circ$



pellet level $\circ$

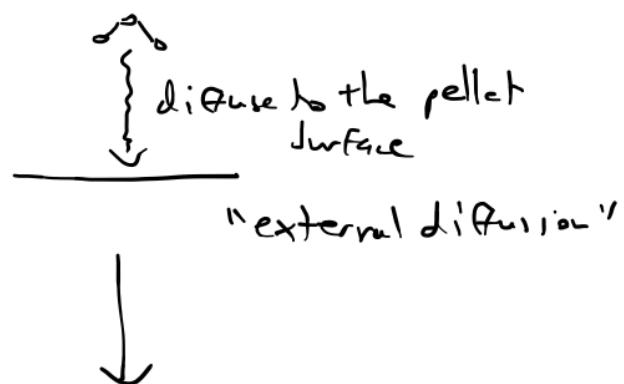
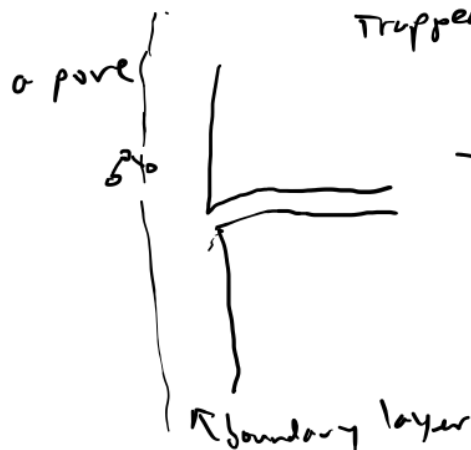


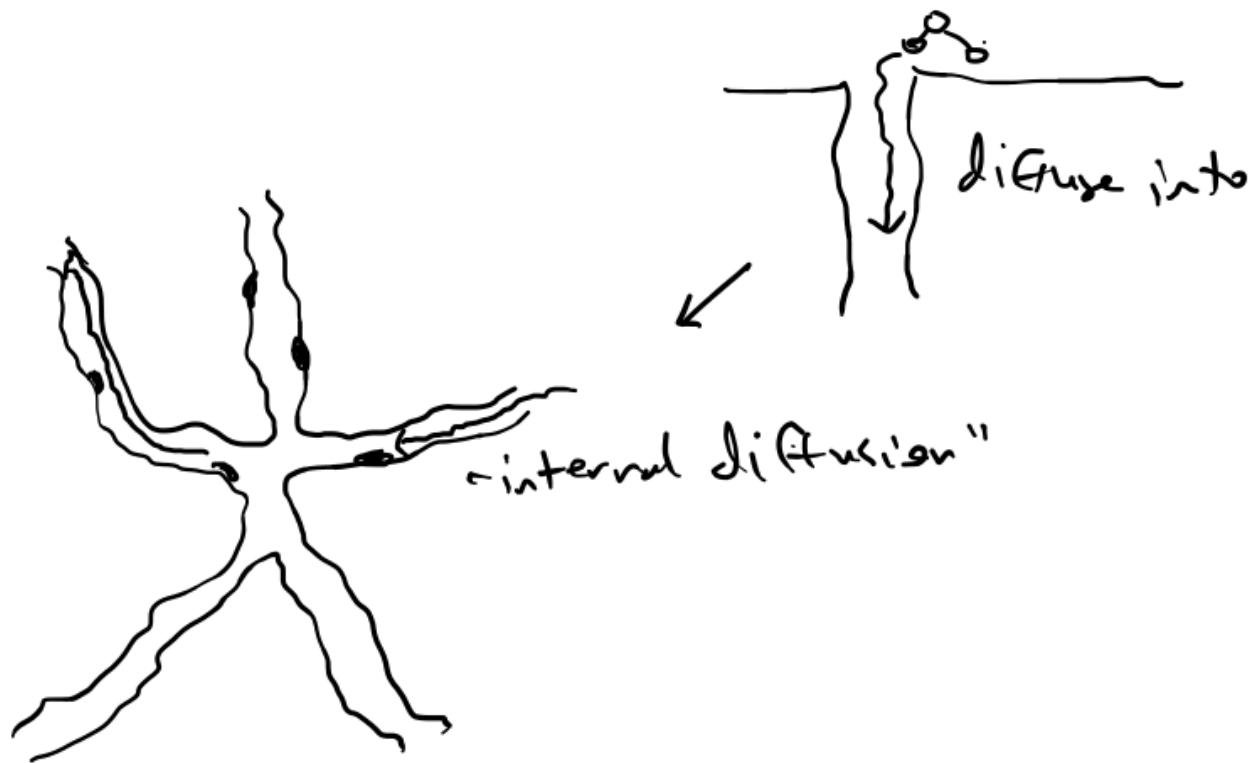
surface from molecule's standpoint

pellet level from pellet standpoint $\circ$



some molecules will be "trapped" by the pellet





- there are several steps in the process of being converted in a catalytic reactor

① external diffusion - reactant & product molecules through boundary layer

② internal diffusion - reactant & product molecules through catalyst pellet pore network

③ catalytic reaction mechanism

④ ANY of these can be the bottleneck ④

- when a reaction is "diffusion limited" - observed rate = rate of diffusion of reactant & product molecules, rate of reaction, $-r_A$, is equal to the rate of diffusion.

$$-r_A = \underset{\substack{\uparrow \\ \text{rate of mass transfer}}}{k_c} (C_{Ab} - \underset{\substack{\uparrow \\ \approx 0}}{C_{Aint}})$$

- Rule of thumb: E^{act} of $20 \frac{\text{kJ}}{\text{mol}}$ or lower, usually suggests diffusion limited reaction.